From +2% to negative real rates

What calibrating the 2-state and 4-state models to Argentina's interest rate tells us

A guide to the claims and findings of notebooks *8. empirical_claims* and *8b. empirical_claims_negative_r*. Giuliana Triberti · draft of October 30, 2026

The question in one paragraph

The thesis argues that treating the informal sector as one low-income group (the **2-state** model: informal, formal) gives a distorted picture compared with splitting it by income (the **4-state** model: IH, IL, FH, FL). This report asks a simple empirical question: *if both models are forced to reproduce the same observed real interest rate r*, does one of them need strange parameter values, or make strange predictions?* The test starts at the advisor's suggested target of r* = +2% and then walks the target down through 0%, −1% and −2% to the medians of Argentina's actual ex-post real rate (−3.56% for 1993–2025 and −5.12% for 1976–2025).

How to read this report

Each section states one **claim** in a grey box with a verdict chip, then gives the evidence: a figure or table, how to read it, and the numbers. Yellow boxes list the **assumptions** each result depends on; their IDs (D = data, C = calibration, N = numerical, J = judgement rule, R = made for this report) point to the full register in Section 6, which also says how each assumption was checked.

The findings at a glance

- **The data do not give +2%.** Argentina's 50-year ex-post real short-term rate is negative (median −5.1%, mean −11.1%). Only the Convertibility years 1991–2001 average about +2% (2.05%). So +2% is a calibration choice, not a data moment.
- **At +2%, the hypothesis that the 2-state model needs a strange parameter is not supported.** Both models reach +2% with ordinary discount rates: ρ = 0.054 (2-state) and ρ = 0.078 (4-state). It is the 4-state model that needs more impatience, and, if credit is the free parameter instead, a much looser borrowing limit (6.6 vs 2.4 months of income).
- **The reason is precautionary saving by formal high-income (FH) workers**, who risk falling to the low-income formal state. It is not the informal high-income (IH) group the thesis emphasises.
- **What calibration pins down is the impatience gap ρ − r*** (≈ 0.034 in the 2-state, ≈ 0.058 in the 4-state), and not ρ or r* separately. Household behaviour barely changes when the target moves, so matching r* alone is a weak test.
- **Toward negative rates, the ranking flips.** Below −2% the 2-state model breaks down: at the data medians it would need a negative discount rate, and no borrowing limit works. The 4-state model reaches −3.56% with ρ = 0.022 and −5.12% with ρ = 0.006.
- **The formalization counterfactual is robust in sign, not in size.** Both models, each at its own calibration, predict that r* rises and fewer households borrow when informality falls. The fall in the borrowers mass in the 4-state model is about one-third of the 2-state fall. The thesis's sign reversal does not survive.
- **Checked against data, the 4-state requirements at +2% are not realistic, and the thesis premium is far too low.** Argentina's household debt has been 4.0%–10.4% of income since 1994 (BIS), against the 17.7% the 4-state borrowing limit implies at +2%. Non-bank lenders charge a real premium of 16–44 pp over banks (BCRA), against θ_l = 2 pp in the thesis. Premia of that size are exactly what the models need to produce the negative data medians.

- **The lesson.** Which model looks 'closer to reality' is decided by the choice of the data counterpart for r*, not by the models. If r* should be a positive real return (the advisor's +2%, or a dollar return), the 4-state model is the one under strain. If it should be the negative ex-post peso rate, only the 4-state model fits. The informal-premium data lean toward the negative reading; the household-debt data favour targets between 0% and −2%.

Target r*	ρ free (in [0.02, 0.12])	a_{min} free (ρ = 0.05)	θ_I free (ρ = 0.05, $\theta_I \geq 0$)
+2% (advisor)	both pass	both pass	neither
0%	both pass	both pass	**2-state only**
−1%	both pass	both pass	both pass
−2%	**4-state only**	**4-state only**	both pass
−3.56% (1993–2025 median)	**4-state only**	**4-state only**	both pass
−5.12% (1976–2025 median)	neither	**4-state only**	**4-state only**

Table 1. Which model can be calibrated to each target with an admissible parameter value. "4-state only" supports the thesis hypothesis that the 4-state model is closer to the data; "2-state only" contradicts it. Pass rules: assumptions J1–J3.

1. The test design

Both models are the continuous-time Huggett economies of the thesis (notebooks 4 and 5), with the full Argentine calibration of Section 2 at η = 0.42 (informal share). Households save or borrow in a single risk-free bond in zero net supply, and r* is the rate that clears that market. Table 2 lists the inputs, copied verbatim from the thesis.

	2-state (I, F)	4-state (IH, IL, FH, FL)
Income y	0.418, 1	0.830, 0.220, 1, 0.321
Tax τ	0, 0.19	0, 0, 0.19, 0.05
Informal borrowing premium θ	0.02, 0	0.02, 0.02, 0, 0
Risk aversion γ	0.30, 0.15	0.30, 0.30, 0.15, 0.15
Transition rates	EPH total hazard 0.914, split to give η	EPH generator (thesis Tables 2–3)
Borrowing limit a_{min}; grid	−0.099; 800 points on [−0.099, 5]	same
Thesis discount rate ρ	0.05	0.05

Table 2. Model inputs (thesis Section 2, η = 0.42).

The test has three rules:

- **One free parameter at a time.** To hit a target r*, free exactly one parameter and keep every other parameter at its thesis value. The main free parameter is the discount rate ρ, which conventionally pins down r* in a Huggett economy. Two parameters that can be checked in data are then freed instead, with ρ held at 0.05: the borrowing limit a_{min} and the informal borrowing premium θ_I.
- **Verify every calibration.** After solving S(r*; parameter) = 0, the full equilibrium is re-solved with the usual bisection over r, and it must return the target.
- **Judge two things:** whether the required parameter value is plausible (assumptions J1–J3), and whether the untargeted predictions look reasonable.

Replication check. Before it is used, the solver reproduces four equilibria printed in the thesis notebooks to within 5×10^{-7} (Table 3), so the results below come from the thesis model itself.

Case (ρ = 0.05)	r* in thesis notebook	|difference|
2-state, Section 2, $\gamma_l = 0.27$, $\eta = 0.42$	0.020229	4.9e-07
2-state, Section 2, $\gamma_l = 0.27$, $\eta = 0.20$	0.033046	2.6e-07
4-state, Section 1 (common τ, γ, $\theta = 0$), $\eta = 0.42$	0.018977	1.1e-07
4-state, Section 2, $\eta = 0.20$	0.010061	1.5e-07

Table 3. The solver reproduces the thesis notebooks (transcribed from notebook 8, Step 2).

Assumptions used here. C1 (only one parameter freed), C2 (thesis inputs, including $\gamma_l = 0.30$ in the 2-state), C4 (r* is the return on the only risk-free asset, in steady state), N1 (same solver and grid as the thesis).

Part I. The advisor's target: r* = +2%

CLAIM 1 · NOTEBOOK 8, STEP 1

NOT SUPPORTED

Argentina's 50-year average real interest rate is about +2%.

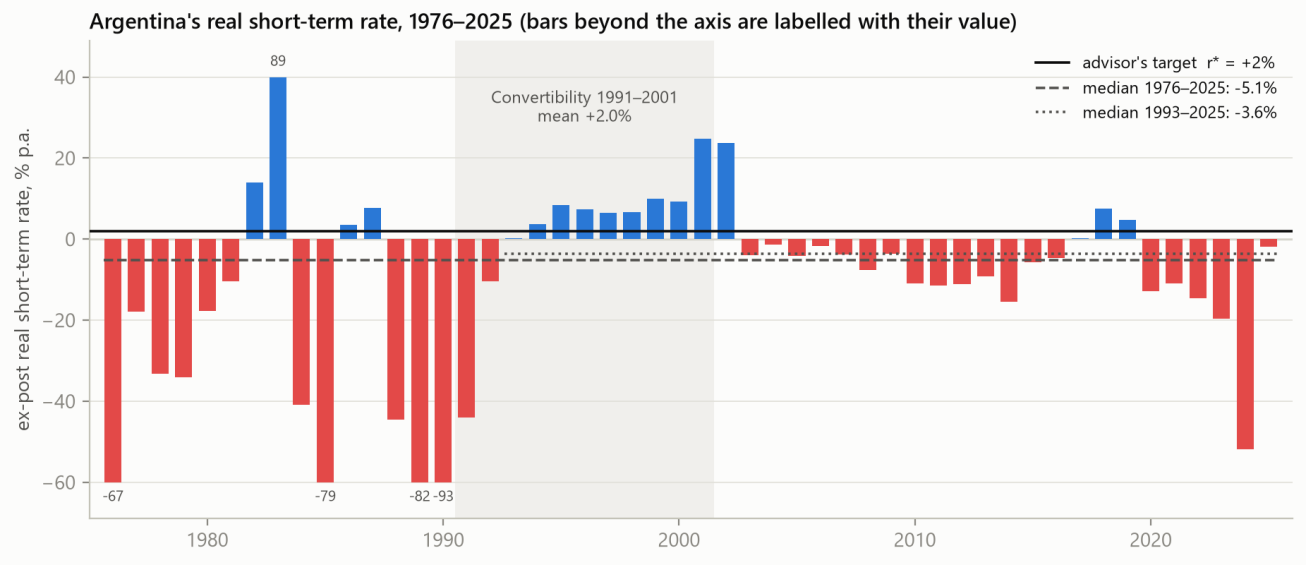


Figure 1. Ex-post real short-term interest rate in Argentina, 1976–2025 (Global Macro Database, release 2026_09). *How to read it:* blue bars are years with a positive real rate and red bars years with a negative one. The solid black line is the +2% target, and the dashed and dotted lines are the medians of the two windows. The grey band marks Convertibility, the only period that averages about +2%.

Window (short-term rate)	Years	Nominal median	Real mean	Real median	Real geom. mean	Years real > 0
1976–2025 (50 years)	50	27.3%	−11.1%	−5.1%	−19.0%	34%
1976–1990 (high inflation)	15	96.7%	−27.0%	−33.2%	−43.4%	27%
1991–2001 (Convertibility)	11	9.1%	2.0%	6.7%	0.3%	82%
2002–2025	24	20.0%	−7.1%	−5.1%	−8.1%	17%
1993–2025	33	12.4%	−2.8%	−3.6%	−3.9%	39%
Policy rate (BIS), 1993–2025	33	9.5%	12.5%	−4.2%	−0.1%	33%

Table 4. Summary of the ex-post real rate by window. Real rates use the exact Fisher relation $(1 + i)/(1 + \pi) - 1$.

What we find

- Every 50-year summary is negative: mean −11.1%, median −5.1%, geometric mean −19.0%. The real rate was positive in only 34% of years.
- Only Convertibility (1991–2001) averages near the target: mean 2.05%, median 6.7%.
- Since 2002 the rate is negative again (median −5.1%). The real policy rate, available only from 1993, has a negative median too.
- **Implication.** +2% is an external calibration choice that matches the Convertibility average, not the 50-year average, and the thesis should say so. Everything in Part I is conditional on it; Part II tests the alternatives the data suggest.

Assumptions used here. D1 (ex-post real rate from realised inflation), D2 (spliced short-term rate as the 50-year series, because Argentina has no continuous policy rate), D3 (GMD release 2026_09, forecasts dropped), C3 (+2% is an external target).

CLAIM 2 · NOTEBOOK 8, STEPS 3–4

FINDING

At the thesis discount rate ρ = 0.05, the two models already disagree about r*: the 2-state model gives +1.59% and the 4-state model −0.76%.

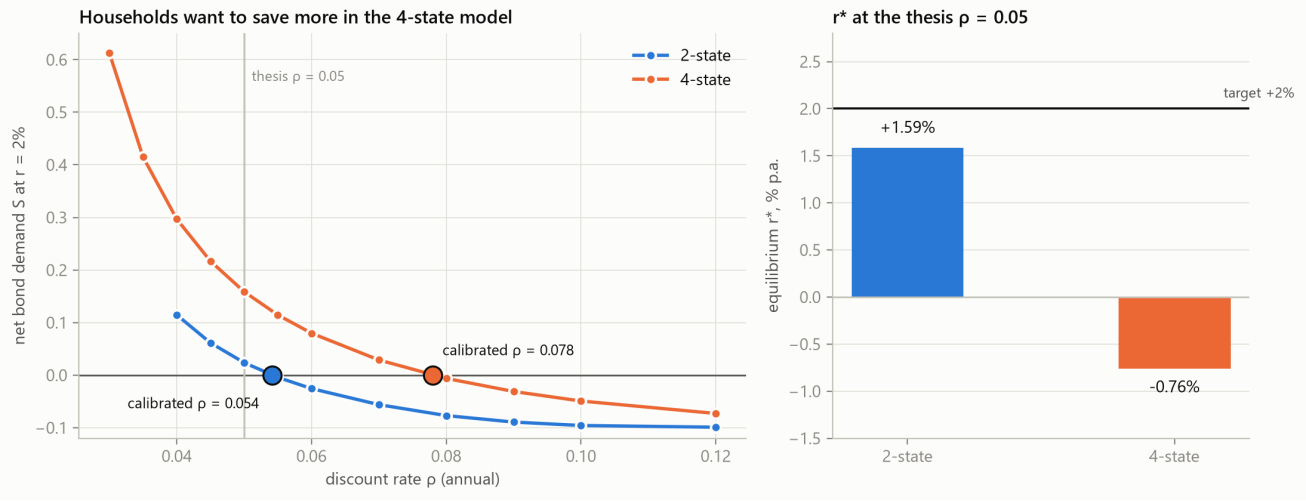


Figure 2. Left: net bond demand S at r = 2% as the discount rate ρ varies. Right: the equilibrium r* each model produces at the thesis ρ = 0.05. *How to read it:* on the left, S > 0 means households want to save more than others want to borrow at 2%, so r* must be lower. Each curve crosses zero at the ρ that makes 2% the equilibrium rate (large dots). The orange curve lies above the blue one everywhere: at any ρ, the 4-state economy wants to hold more bonds.

- At ρ = 0.05 and r = 2%, the 4-state economy has excess bond demand 0.159, against 0.024 in the 2-state model, about 7 times as large.
- So with the thesis parameters the 4-state model needs a much lower rate to clear the market. It clears **below zero** (−0.76%). This is the first hint that the 4-state model fits low or negative rates more naturally.

Assumptions used here. C2 (thesis inputs), N1 (solver). The lower bound on r used in the thesis notebooks was removed here so that the true market-clearing rate is reported (see Claim 6 for why this matters).

CLAIM 3 · NOTEBOOK 8, STEPS 4–5 AND 8

NOT SUPPORTED

The 2-state model needs a strange (implausible) discount rate to produce r* = +2%, while the 4-state model does not.

Model	Calibrated ρ	Annual β = $e^{-\rho}$	Quarterly β	Impatience gap ρ − r*
2-state	0.0542	0.947	0.9866	0.0342
4-state	0.0780	0.925	0.9807	0.0580

Table 5. Discount rate that makes r* = +2% in each model (verified: the re-solved equilibrium returns 2.0000% in both).

Target r* (robustness)	ρ, 2-state	ρ, 4-state	$\rho_{4s} - \rho_{2s}$
1.0%	0.0441	0.0678	0.0238
1.5%	0.0491	0.0729	0.0238
2.0%	0.0542	0.0780	0.0238
2.5%	0.0592	0.0830	0.0238
+2%, 2-state with $\gamma_I = 0.27$	0.0498		
+2%, calibrated at η = 0.20	0.0393	0.0600	0.0207

Table 6. The gap between the models does not depend on the 2% choice, on γ_I, or on the size of the informal sector (last two rows transcribed from notebook 8, Step 8).

What we find

- 2-state: ρ = 0.054 (annual β = 0.947). That is a textbook value; at the thesis ρ = 0.05 it already gives r* = 1.6%.
- 4-state: ρ = 0.078 (annual β = 0.925, quarterly 0.981). This is more impatient, but it is the standard value in emerging-market business-cycle models (quarterly β ≈ 0.98, as in Aguiar and Gopinath 2007 and Neumeyer and Perri 2005).
- The gap $\rho_{4s} - \rho_{2s} = 0.024$ is the same for every target from 1% to 2.5%.
- **Verdict.** Neither value is strange, so neither model is ruled out on parameter grounds. If anything, the 4-state model is the one that needs the less conventional value.

Assumptions used here. J1 (a value of ρ is 'plausible' if it lies in [0.02, 0.12] per year), C2 ($\gamma_I = 0.30$; the notebook-4 code uses 0.27, tested in Table 6), C1.

CLAIM 4 · NOTEBOOK 8, STEP 9

NOT SUPPORTED

The 4-state model saves more because of the informal high-income (IH) group.

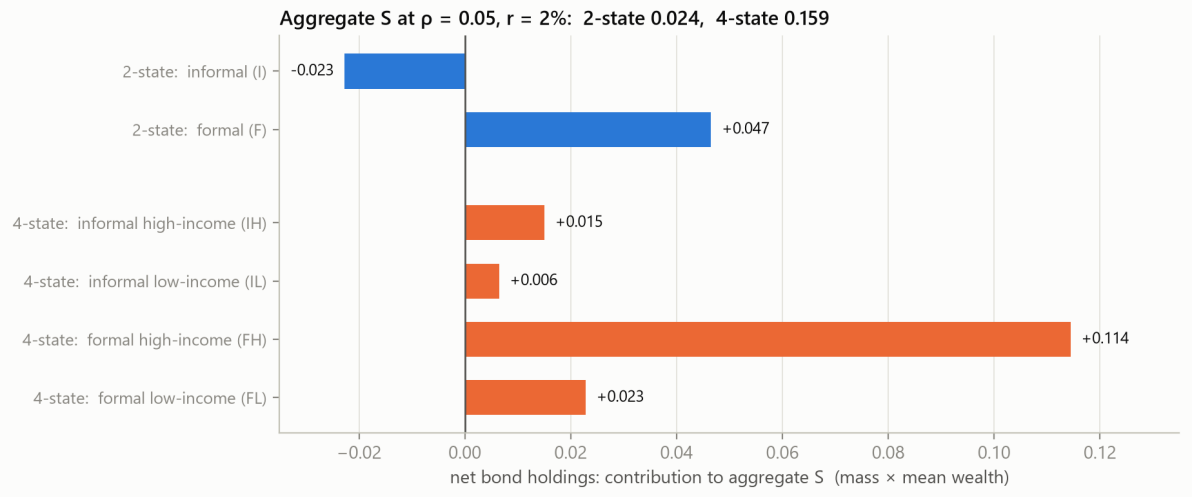


Figure 3. Each group's contribution to aggregate bond holdings S at a common ρ = 0.05 and r = 2%. *How to read it:* the bar is the group's mass times its mean wealth, and the bars of a model add up to its aggregate S (in the title). Bars to the right of zero are net savers. The longest bar, formal high-income (FH) workers in the 4-state model, is where the extra saving comes from.

Model	State	After-tax income	Mass	Mean wealth	Contribution to S	Exit rate per year
2-state	I	0.418	0.420	−0.054	−0.0228	0.530
2-state	F	0.810	0.580	0.080	0.0465	0.384
4-state	IH	0.830	0.077	0.195	0.0150	3.154
4-state	IL	0.220	0.343	0.019	0.0065	0.984
4-state	FH	0.810	0.403	0.284	0.1144	0.909
4-state	FL	0.305	0.177	0.129	0.0228	2.784

Table 7. Decomposition behind Figure 3 (ρ = 0.05, r = 2%).

- FH workers hold 0.114 of net bonds, against 0.047 for the whole formal group in the 2-state model. IH workers contribute only 0.015.
- FH workers leave their state at a rate of 0.91 per year, and 0.68 of that (75%) is a fall to low-income formal work (income 0.32 instead of 1). Facing a large, frequent income drop, they build precautionary savings. The 2-state model averages this risk away inside a single formal state.
- **Implication.** The thesis should not claim that IH workers are what depresses r*. The income risk *within* the formal sector, which the 4-state model also captures, is the driver.

Assumptions used here. R3 (reading the exit rates as income risk is an interpretation of the generator, not a separate estimate), C2.

CLAIM 5 · NOTEBOOK 8, STEP 6

NOT SUPPORTED

Once both models are calibrated to +2%, one of them makes implausible predictions.

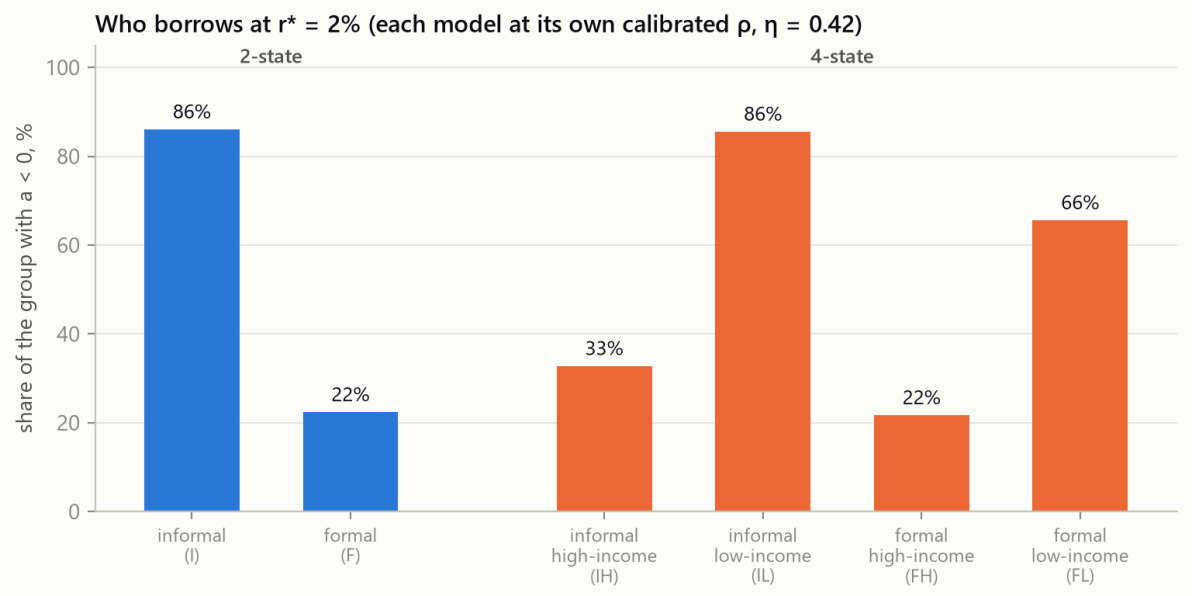


Figure 4. Share of each group in debt (a < 0), with each model at its own calibrated ρ and r* = +2%. *How to read it:* bars are grouped by model. In both models informal workers borrow more than formal workers. The 4-state model adds that low-income formal workers (FL) mostly borrow, while IH workers mostly save.

Prediction (η = 0.42, r* = 2%)	2-state	4-state
Borrowers mass (share with a < 0)	0.491	0.522
Mass at the borrowing limit	0.234	0.219
Informal: share borrowing	0.860	0.758
Formal: share borrowing	0.224	0.351
Informal mean wealth	−0.066	−0.041
Formal mean wealth	0.048	0.030
Gross debt / mean after-tax income	0.056	0.071

Table 8. Untargeted moments at the common calibration.

- The aggregate predictions are close: about half the population borrows in both models, and just over a fifth sits at the borrowing limit.
- Neither model predicts anything implausible. Wealth and borrowing are not observed in the EPH, so these moments cannot tell the models apart with the thesis data. They are the natural place to bring household-finance data later (Section 5).

CLAIM 6 · NOTEBOOK 8, STEP 7 (THESIS SECTION 4.4) **NOT SUPPORTED**

Formalization moves the borrowers mass in opposite directions in the two models (the thesis's 'sign reversal').

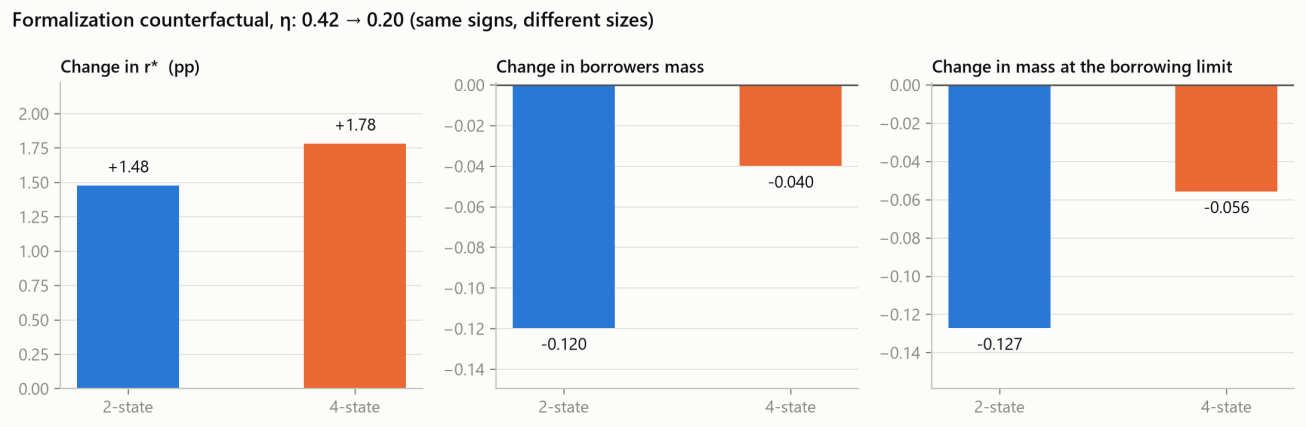


Figure 5. Effect of lowering informality from η = 0.42 to η = 0.20, with each model at its own calibrated ρ. *How to read it:* each panel is one outcome and the bars show the change in each model. Bars on the same side of zero mean that the models agree on the direction; different heights mean that they disagree on the size.

Model	ρ	η	r*	Borrowers mass	Mass at limit	Informal borrowing	Formal borrowing
2-state	0.0542	0.42	2.00%	0.491	0.234	0.860	0.224
2-state	0.0542	0.20	3.48%	0.371	0.107	0.901	0.239
4-state	0.0780	0.42	2.00%	0.522	0.219	0.758	0.351
4-state	0.0780	0.20	3.78%	0.482	0.164	0.803	0.402

Table 9. Formalization counterfactual; every equilibrium cleared.

- **Same signs.** r* rises (+1.48 pp in the 2-state, +1.78 pp in the 4-state) and the borrowers mass falls in both.

- **Different sizes.** The borrowers mass falls by −0.120 in the 2-state model but only −0.040 in the 4-state, about 33% of it. The mass at the constraint falls by −0.127 against −0.056.
- **Why the thesis found a reversal.** In thesis notebook 5, the 4-state equilibrium at η = 0.42 and ρ = 0.05 never cleared: the bisection stopped at its lower bound r = 0.0005 with S = +0.03. That point is not an equilibrium. Claim 2 shows the true r* there is negative.
- **What survives.** The 4-state model implies a much weaker response of financial constraints to formalization. That difference in size is policy-relevant, and it does not depend on how r* is calibrated.

Assumptions used here. C1, C2; ρ held at each model's own calibrated value while η changes (the counterfactual keeps preferences fixed). Only η = 0.20 is tested.

CLAIM 7 · NOTEBOOK 8, STEPS 11–12

NOT SUPPORTED

With parameters that can be checked in data (credit limit, informal premium) instead of ρ, the 4-state model matches +2% more easily.

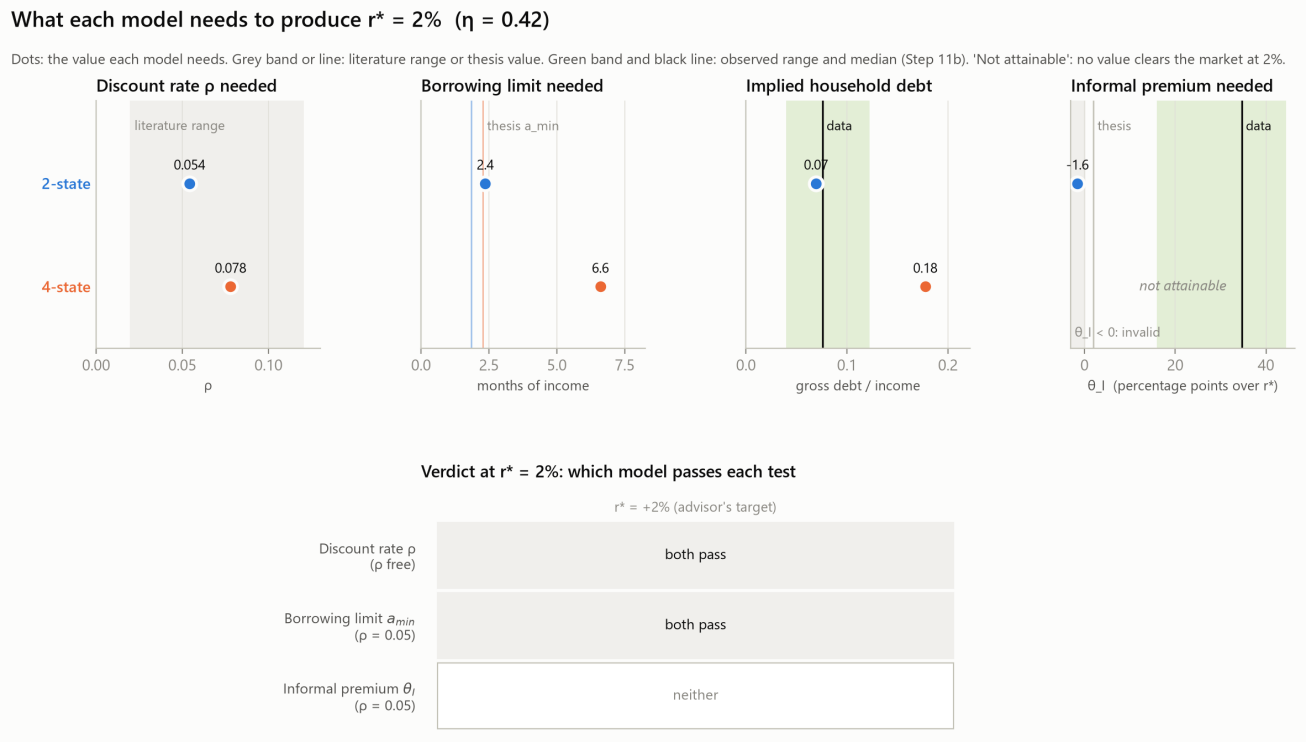


Figure 6. What each model needs to produce r* = +2% when ρ is held at 0.05 and one observable parameter is freed, with the pass/fail verdict. *How to read it:* dots show the value each model needs. Grey bands and thin lines mark the literature range or the thesis value. 'Not attainable' means that no value of the parameter clears the market at 2%. The lower panel applies the pass rules.

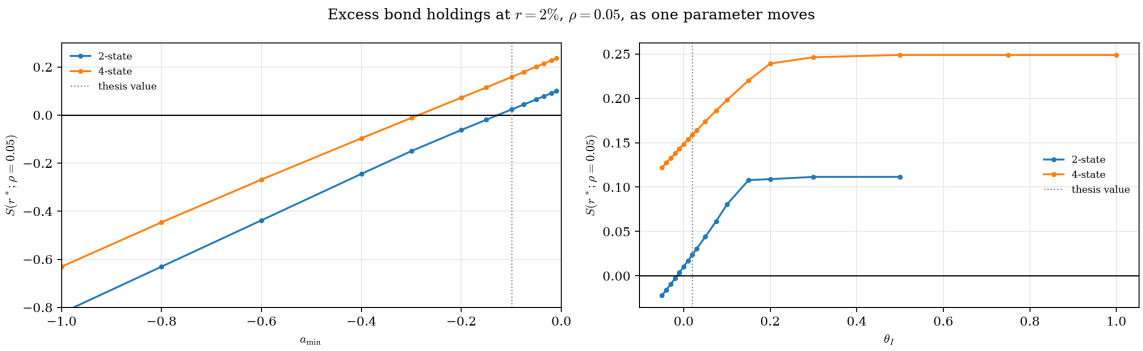


Figure 7. Supporting evidence: net bond demand at r = 2% and ρ = 0.05 as a_{min} (left) or θ_l (right) varies. *How to read it:* a calibration exists where a curve crosses zero. The 4-state curve on the right stays above zero for every premium from −5% to +100%.

- **Borrowing limit.** 2-state: a_{min} = −0.127 (2.4 months of income; thesis value about 1.8 months), debt/income 0.069. 4-state: a_{min} = −0.286 (6.6 months), debt/income 0.177, 2.6 times as much.
- **Informal premium.** 2-state: θ_l = −0.016, a *negative* premium, so informal households would borrow below the saving rate; that is not economically meaningful. 4-state: not attainable. Even nearly free informal credit cannot absorb the excess saving, because that saving comes from FH households (Claim 4), whom θ_l does not reach.
- **Verdict at +2%.** No test is passed by the 4-state model and failed by the 2-state model. Through credit, the 4-state model needs household debt of 17.7% of income. Part III shows that Argentina's has been 4.0%–10.4% since 1994, so the 2-state requirement (6.9%) is the realistic one (green bands in Figure 6).

Assumptions used here. C5 (thesis a_{min} = −0.099 has no stated source; θ_l = 0.02), J2 (θ_l < 0 fails), J3 (the pass rule only asks that a_{min} exists and clears; its realism is judged separately in Part III), N4 ('not attainable' is defined on the scan grid), D5–D8 (data benchmarks drawn in the figure).

Part II. Walking the target down to the data

Part I took +2% as given. But Claim 1 showed that Argentina's real rate is negative, and Claim 2 showed that the 4-state model clears below zero at the thesis parameters. Notebook 8b therefore repeats every test for targets of 0%, −1%, −2%, the 1993–2025 median (−3.56%) and the 1976–2025 median (−5.12%).

CLAIM 8 · NOTEBOOK 8B, STEPS 2–4

NOT SUPPORTED

Matching r* identifies the discount rate.

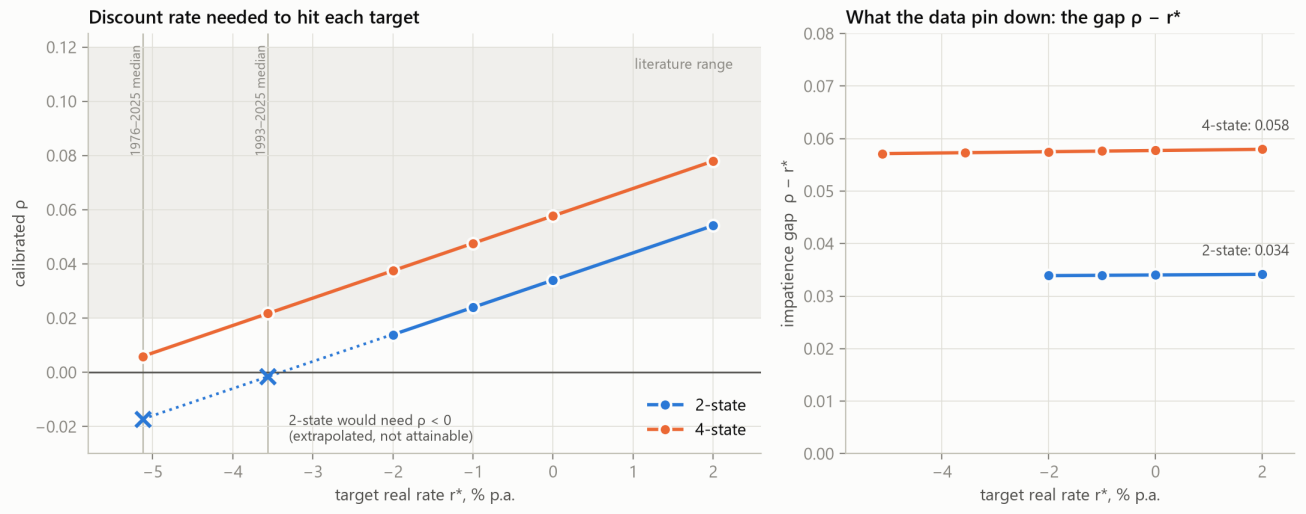


Figure 8. Left: the ρ needed to hit each target. Right: the impatience gap ρ − r* at each calibration. *How to read it:* on the left, the grey band is the literature range for ρ and the vertical lines are the two data medians. Crosses mark targets where the 2-state model cannot be calibrated with ρ > 0; the dotted line extrapolates its constant gap. On the right, both lines are flat: each model needs the same ρ − r* whatever the target.

Target r*	ρ, 2-state	ρ, 4-state	ρ − r*, 2-state	ρ − r*, 4-state
+2% (advisor)	0.0542	0.0780	0.0342	0.0580
0%	0.0340	0.0577	0.0340	0.0577
−1%	0.0240	0.0476	0.0340	0.0576
−2%	0.0139	0.0375	0.0339	0.0575
−3.56% (1993–2025 median)	–	0.0217	–	0.0573
−5.12% (1976–2025 median)	–	0.0059	–	0.0571

Table 10. Calibrated ρ and impatience gap by target ('–' means no ρ > 0 reaches the target).

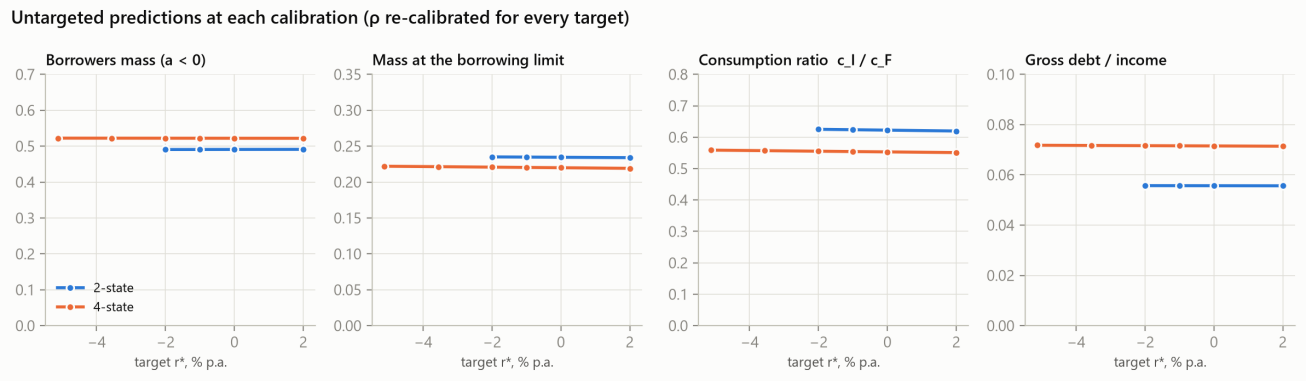


Figure 9. Untargeted predictions at each calibration, with ρ re-calibrated for every target. *How to read it:* every y-axis starts at zero, so a flat line means the prediction does not depend on the target. All four predictions are essentially flat in both models.

- The calibration pins down the gap ρ − r* (about 0.034 in the 2-state and 0.058 in the 4-state model), not the level of either. Shifting r* and ρ together leaves household behaviour almost unchanged.
- **Implication.** Hitting a target r* is a weak test of a model: any target can be matched by moving ρ in step, and the predictions barely move. What discriminates is whether the required ρ is admissible (positive, inside the literature range) and what the observable parameters must be.

Assumptions used here. C4, N2 (for r ≤ 0 the solver starts from a different initial guess and uses continuation; the fixed point is unchanged, checked at r = 2% to 3 × 10^{-10}), R2 (the dotted extrapolation assumes the 2-state gap stays constant).

CLAIM 9 · NOTEBOOK 8B, STEPS 2 AND 7

SUPPORTED

At Argentina's ex-post real-rate medians, only the 4-state model can be calibrated with a positive discount rate.

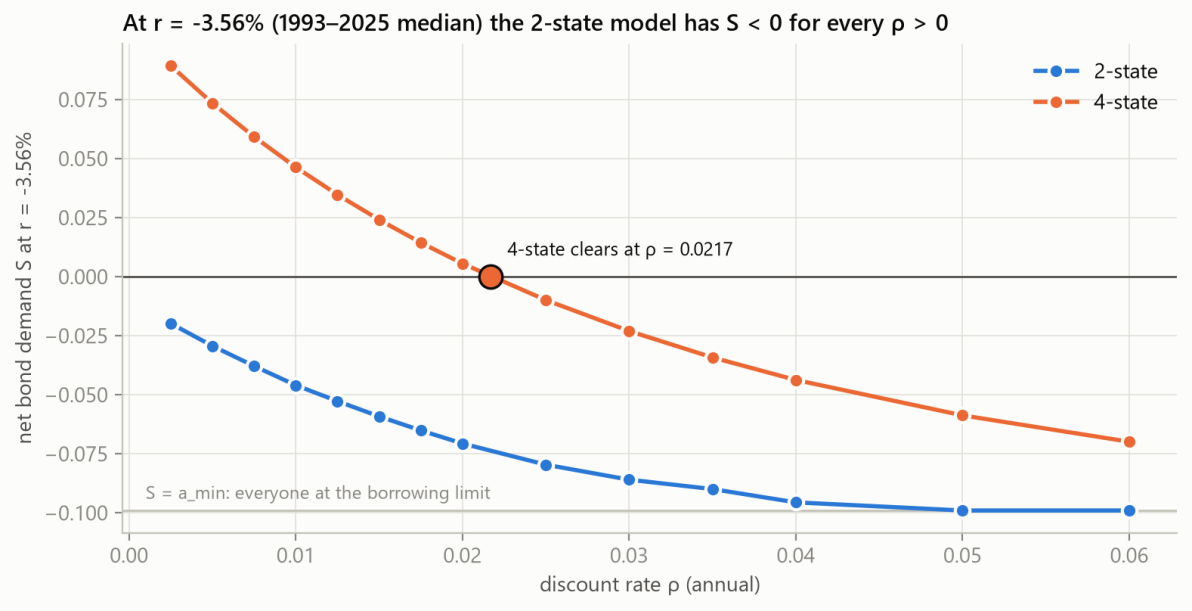


Figure 10. Net bond demand at r = −3.56% (the 1993–2025 median) as ρ falls toward zero. *How to read it:* a calibration exists where a curve crosses zero. The 4-state curve crosses at ρ ≈ 0.02. The 2-state curve stays below zero all the way down to ρ = 0.0025: even very patient households want to borrow more than others want to lend. At higher ρ it sits at the corner S = a_{min}, where everyone is at the borrowing limit.

- **4-state:** reaches −3.56% with ρ = 0.0217 (annual β = 0.979), a textbook value, and −5.12% with ρ = 0.0059, which is very patient but positive.
- **2-state:** S < 0 at every ρ tried, down to 0.0025. Extrapolating its constant gap, it would need ρ ≈ −0.2% and −1.7%: households that value the future *more* than the present, which is not a standard parameter.
- **Why.** The 4-state model carries more precautionary saving (Claims 2 and 4). At the same ρ it wants to hold more bonds, so only a low or negative rate clears the market. That same feature made it strain at +2%.

Assumptions used here. D4 (the medians, not the means, are the data targets: the means are dominated by the 1975–1991 high inflation), J1, R2 (the negative-ρ figures are extrapolations, not solves).

CLAIM 10 · NOTEBOOK 8B, STEPS 5–6

SUPPORTED

The ranking of the models on observable parameters also flips as the target turns negative.

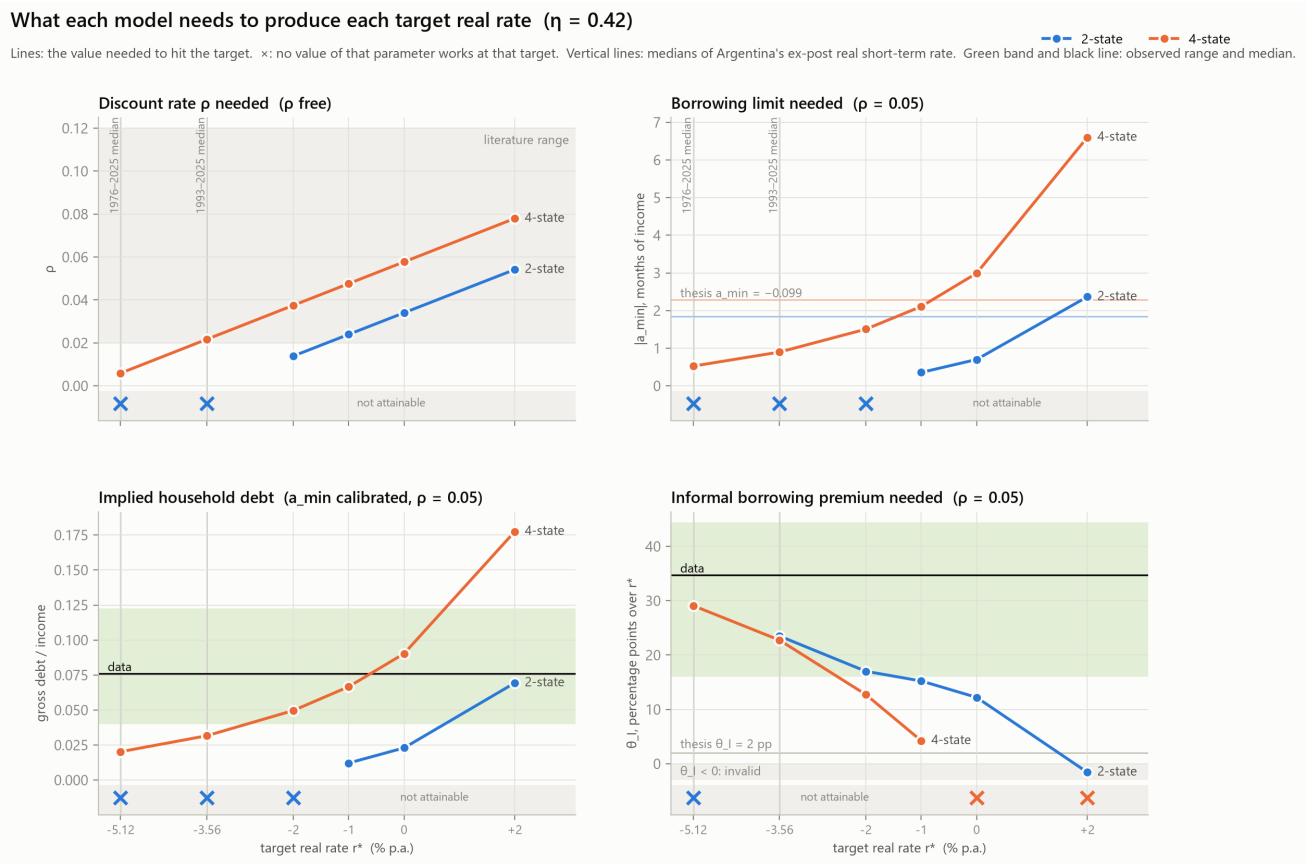


Figure 11. What each model needs to hit each target: ρ (top left), and, with ρ = 0.05, the borrowing limit (top right), the implied household debt (bottom left) and the informal premium (bottom right). *How to read it:* lines show the value needed, and crosses in the grey strip at the bottom mark targets where no value works. Follow each line from right (+2%) to left (the data medians). The blue 2-state line ends early in three of the four panels.

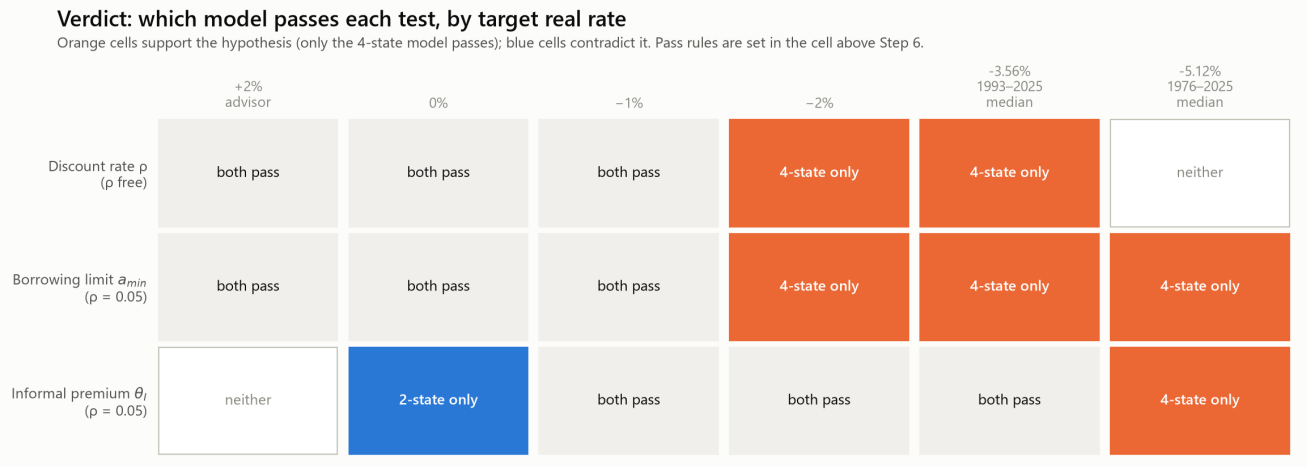


Figure 12. Pass/fail verdict for each test and target (same pass rules as Table 1). *How to read it:* orange cells (4-state only) support the hypothesis that the 4-state model is closer to the data; blue cells contradict it. They concentrate at the negative targets.

Target r*	a_{min}, 2-state	a_{min}, 4-state	θ_l, 2-state	θ_l, 4-state
+2% (advisor)	−0.127 (2.4 mo)	−0.286 (6.6 mo)	−1.6 pp	not attainable
0%	−0.038 (0.7 mo)	−0.130 (3.0 mo)	12.2 pp	not attainable
−1%	−0.019 (0.4 mo)	−0.091 (2.1 mo)	15.2 pp	4.2 pp
−2%	not attainable	−0.065 (1.5 mo)	17.0 pp	12.7 pp
−3.56% (1993–2025 median)	not attainable	−0.039 (0.9 mo)	23.5 pp	22.7 pp
−5.12% (1976–2025 median)	not attainable	−0.023 (0.5 mo)	no robust eq.	29.1 pp

Table 11. Observable parameter needed at ρ = 0.05 (a_{min} also in months of mean after-tax income; θ_l in percentage points over r*).

- **Positive targets:** the 4-state model is the one needing an implausible observable parameter: a very loose borrowing limit, and no informal premium works at all at 0% or +2%.
- **Negative targets:** the 2-state model cannot reach −2% or below through the borrowing limit. Even at a_{min} = −0.01 almost everyone sits at the limit and S stays negative. Through θ_l it needs high premia, and at the 1976–2025 median the fixed-r root does not survive re-solving (no robust equilibrium).
- The 4-state model reaches every negative target with a tight but positive borrowing limit (about 0.9 months of income at −3.56%), or with a real informal borrowing rate r + θ_l of about 19%.
- These a_{min} values are not grid artefacts: re-calibrating on a grid three times finer moves them by less than 4%, except the 2-state model at −1% (8%, with only 4 grid points below zero). See Table 12.

Target	Model	Grid points below 0 (I = 800)	a_{min}, I = 800	a_{min}, I = 2400	Change
+2%	2-state	20	−0.1271	−0.1274	+0.2%
+2%	4-state	44	−0.2860	−0.2846	−0.5%
0%	2-state	6	−0.0377	−0.0390	+3.4%
0%	4-state	21	−0.1296	−0.1294	−0.2%
−1%	2-state	4	−0.0191	−0.0206	+7.9%
−1%	4-state	15	−0.0913	−0.0915	+0.2%
−2%	4-state	11	−0.0653	−0.0655	+0.3%
−3.56%	4-state	7	−0.0388	−0.0392	+1.0%
−5.12%	4-state	4	−0.0227	−0.0235	+3.5%

Table 12. Grid-resolution check for tight borrowing limits (transcribed from notebook 8b, Step 5).

Assumptions used here. J2, J3, N4, N5 (grid check), N6 (the 2-state θ_l root at −5.12% fails to clear when re-solved and is treated as not attainable).

Part III. Are the required parameters realistic? A check against data

The pass rules of Parts I and II only ask whether a calibration exists and is economically meaningful. This part asks whether the borrowing limit and the informal premium each model needs are the size we observe in Argentina. Notebook 8, Step 11b, builds two benchmarks from public data (Table 13). The raw files and their sources are in *data/interest_rates/benchmarks/*.

- **Household gross debt / income**, the benchmark for a_{min} through the debt it implies: BIS credit to households (% of GDP) divided by household consumption (% of GDP, World Bank). In the model, income is after-tax income, and with bonds in zero net supply aggregate consumption equals it, so consumption is the model-consistent denominator. An upper series adds non-bank lenders, which the BCRA puts at 18% of bank credit to individuals.
- **Informal borrowing premium**, the benchmark for θ_I. In the model, formal households borrow at r and informal ones at $r + \theta_I$, so θ_I is the gap in borrowing cost between the two groups. The data counterpart is the personal-loan rate of non-bank credit providers (fintech, retailers, cooperatives), who lend to people without access to bank credit, over the bank personal-loan rate. The real spread is $(1 + i_{non\text{-}bank})/(1 + i_{bank}) - 1$, which does not depend on the inflation measure.

Benchmark	Central value	Observed range	Source
Household debt / income	7.6% (1994–2025 median)	4.0%–10.4% (BIS); up to 12.3% with non-bank lenders	BIS total credit statistics (Q.AR.H.A.M.770.A); World Bank WDI (NE.CON.PRVT.ZS); BCRA (2026)
Latest year	7.8% (2025)		
Convertibility, 1994–2001 (the +2% years)	7.7% (mean)	peak 10.4% (2000)	
Informal premium θ_I	35 pp (median of 3 dates)	16 pp (cooperatives) to 44 pp (all providers, Feb 2026)	BCRA, *Informe de Proveedores No Financieros de Crédito*, Dec 2023, Nov 2025, Jun 2026
Thesis value of θ_I	2 pp		Thesis Section 2

Table 13. Data benchmarks (notebook 8, Step 11b). The green bands in the figures are the observed ranges.

Table 14 summarises the result of Claims 11 and 12 below. Feasibility (Table 1) asks whether a value exists; realism asks whether that value is the size observed in Argentina.

Target r*	a_{min} route: implied debt realistic	θ_I route: premium realistic
+2% (advisor)	**2-state only**	neither
0%	**4-state only**	neither
−1%	**4-state only**	neither
−2%	**4-state only**	**2-state only**
−3.56% (1993–2025 median)	neither	both
−5.12% (1976–2025 median)	neither	**4-state only**

Table 14. Realism at a glance: which model's requirement lies inside the observed range (J4). Compare with the feasibility verdict of Table 1.

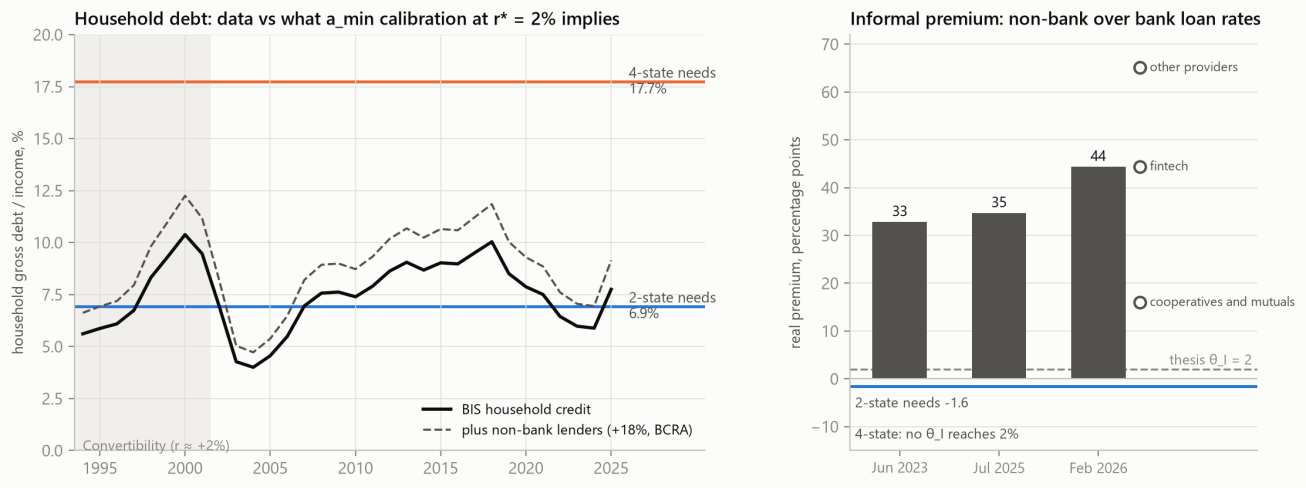


Figure 13. Left: Argentina's household debt / income, 1994–2025, against the debt each model's borrowing limit implies at r* = +2% (ρ = 0.05). Right: real premium of non-bank over bank personal-loan rates, against the thesis value and what each model needs at +2%. *How to read it:* on the left, the black line is the data and the dashed line adds non-bank lenders. The orange line (4-state requirement) lies above the data in every year, including the Convertibility peak; the blue line (2-state) runs through the data. On the right, the bars are the all-provider premium at three dates and the open dots are provider groups in February 2026. Both the thesis 2 pp and the premia the models need at +2% sit at the bottom of the chart.

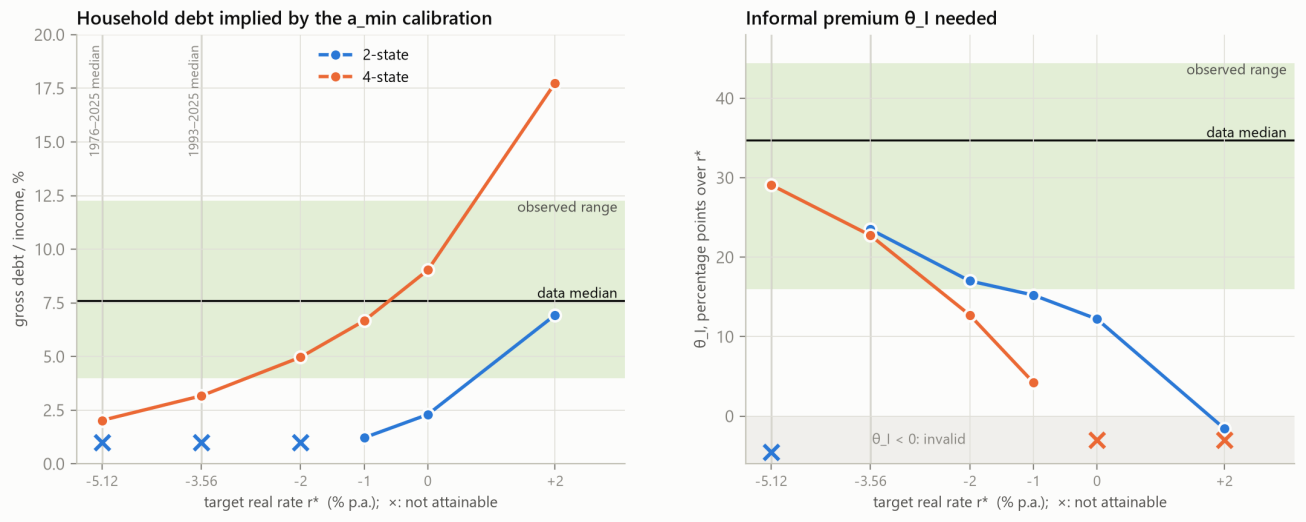


Figure 14. What each model needs at each target, against the observed range (green band) and median (black line): household debt implied by the a_{min} calibration (left) and informal premium (right), ρ = 0.05. *How to read it:* follow each line from right (+2%) to left (the data medians) and check when it is inside the green band. Crosses at the bottom mark targets where no value works. The left panel favours moderate targets; the right panel favours negative ones.

CLAIM 11 · NOTEBOOK 8, STEPS 11–12 AND 11B; NOTEBOOK 8B, STEP 6

DEPENDS ON THE TARGET

The household debt implied by the 4-state model's borrowing limit is consistent with Argentina's.

Model at r* = +2%	a_{min}	Limit, months of mean income	Limit, months of own income (poorest group)	Share in debt	Share at the limit	Debt per borrower, months of mean income	Debt / income
2-state	−0.127	2.4	3.6 (I)	49%	21%	1.7	6.9%
4-state	−0.286	6.6	15.6 (IL)	52%	10%	4.1	17.7%
Data							4.0%–12.3%

Table 15. What the borrowing limit needed at +2% means for households (a_{min} calibrated, ρ = 0.05). The limit is the most anyone may owe, not what people owe: debt per borrower is total debt divided by the share in debt.

Target r*	Debt / income, 2-state	Debt / income, 4-state	Inside the data range (4.0%–12.3%)
+2% (advisor)	6.9%	17.7%	**2-state only**
0%	2.3%	9.0%	**4-state only**
−1%	1.2%	6.7%	**4-state only**
−2%	not attainable	5.0%	**4-state only**
−3.56% (1993–2025 median)	not attainable	3.2%	neither
−5.12% (1976–2025 median)	not attainable	2.0%	neither

Table 16. Household debt implied by the a_{min} calibration at each target, against the data.

- **What the 6.6 months means.** a_{min} is the most a household may owe. At +2% the 4-state limit is 6.6 months of *average* income, which is 15.6 months of a low-income informal worker's own income. Only 10% of households are at the limit; the average borrower owes about 4.1 months of average income.
- **At +2% the 4-state requirement is not realistic.** It implies debt of 17.7% of income: 2.3 times the 1994–2025 median, above every year on record (bank peak 10.4%), and above even the upper series with non-bank lenders (12.3%). The 2-state requirement (6.9%) is close to the median.
- **Between 0% and −2%, only the 4-state requirement is realistic.** The 2-state limit has to become so tight that debt falls to 1–2% of income, and below −1% no limit works.
- **At the data medians, the 4-state requirement is too tight.** It implies debt of 3.2% and 2.0% of income, below the lowest year on record (4.0%, 2004). On this check, the a_{min} route fits the medians in neither model; the ρ route of Claim 9 is the one that works there.

Assumptions used here. D5 (BIS household credit as the debt numerator), D6 (household consumption as the income denominator), J4 (a requirement is 'realistic' if it lies inside the observed range), C1 (ρ = 0.05 and all other parameters at thesis values while a_{min} moves).

CLAIM 12 · NOTEBOOK 8, STEP 11B; NOTEBOOK 8B, STEPS 5–6

NOT SUPPORTED

The informal borrowing premium observed in Argentina is compatible with r* = +2%.

Date	Lender	Non-bank rate (TNA)	Bank rate (TNA)	Real premium	BCRA source
Jun 2023	all non-bank providers	171%	104%	32.8 pp	Report of Dec 2023, p. 11
Jul 2025	all non-bank providers	129%	70%	34.7 pp	Report of Nov 2025, p. 14
Feb 2026	all non-bank providers	144%	69%	44.4 pp	Report of Jun 2026, p. 13
Feb 2026	cooperatives and mutuals	96%	69%	16.0 pp	Report of Jun 2026, p. 14
Feb 2026	fintech	144%	69%	44.4 pp	Report of Jun 2026, p. 14
Feb 2026	other providers	179%	69%	65.1 pp	Report of Jun 2026, p. 14

Table 17. Personal-loan rates of non-bank credit providers and banks (nominal annual rates), and the implied real premium $(1 + i_{non\text{-}bank})/(1 + i_{bank}) - 1$. Source: BCRA, *Informe de Proveedores No Financieros de Crédito*.

Target r*	θ_l needed, 2-state	θ_l needed, 4-state	Inside the data range (16–44 pp)
+2% (advisor)	−1.6 pp	not attainable	neither
0%	12.2 pp	not attainable	neither
−1%	15.2 pp	4.2 pp	neither
−2%	17.0 pp	12.7 pp	**2-state only**
−3.56% (1993–2025 median)	23.5 pp	22.7 pp	both
−5.12% (1976–2025 median)	no robust eq.	29.1 pp	**4-state only**

Table 18. Informal premium needed at each target (ρ = 0.05), against the data.

- **The observed premium is large.** Non-bank lenders charge a real premium of 33–44 pp over banks, and 16 pp for the cheapest group (cooperatives and mutuals). The thesis value of 2 pp is far below every observation.
- **Papers agree on the order of magnitude.** In its survey of informal credit markets, Batini, Kim, Levine and Lotti (2010, IMF WP/10/42) report that informal lenders' rates are on average much higher than formal rates, range from 0 to 200% a year, and often exceed 75% a year (citing Besley 1994 and Hoff and Stiglitz 1990). The BCRA links rate differences to business models, funding costs and client risk; non-bank clients are riskier: 34% of their personal loans were irregular in February 2026, against 14% at banks.
- **A premium of that size is incompatible with +2% or 0%.** At +2% the 2-state model needs a negative premium and the 4-state model none works; at 0% the 4-state model still has no solution and the 2-state needs 12 pp, below the range.
- **It is the size that produces a negative r*.** At the 1993–2025 median both models need about 23 pp, and at the 1976–2025 median the 4-state model needs 29 pp. Both are inside the observed range. This is the one data check that favours negative targets whatever one thinks of the ex-post peso rate as a measure of r*.
- **Caveat.** The premium check does not separate the models at the 1993–2025 median (both pass); it separates them only at the 1976–2025 median, where the 2-state root does not survive re-solving.

Assumptions used here. D7 (non-bank over bank rates as the informal premium), D8 (ratio of nominal rates as the real premium; rates are TNA, not effective; the spread includes default compensation), J4, C1 (ρ = 0.05 and thesis a_{min} while θ_l moves).

Part IV. What we learn

1. The +2% target does not favour the 4-state model, and the data confirm it

At +2% both models calibrate with admissible discount rates, but the 4-state model needs more impatience (ρ = 0.078 vs 0.054), a borrowing limit of about 6.6 months of income, and household debt of 17.7% of income, above every year in the BIS data (Claim 11). No informal premium can make it clear at 2%, and the premia observed in Argentina are far larger than any value compatible with +2% (Claim 12). The 2-state model needs only modest changes. On this target, the claim that the 2-state model needs strange parameters is not supported.

2. Matching r* is a weak test, because only ρ − r* is pinned down

Moving the target moves ρ one-for-one and leaves every household prediction unchanged (Claim 8). A model cannot be validated or rejected by matching r* alone. The informative content is (a) whether the required ρ is admissible and (b) what the observable parameters, borrowing limits and premia, must be to fit. Those can be checked against data.

3. The 4-state model's extra precautionary saving is what lets it fit negative rates

Income risk inside the formal sector (FH ↔ FL) makes 4-state households want to save more at any rate (Claims 2 and 4). That is a weakness at +2% and a strength at negative rates. At the medians of the actual ex-post real rate, only the 4-state model can be calibrated with a positive discount rate and an admissible borrowing limit (Claims 9 and 10).

4. So the decisive question is what r* corresponds to in Argentina

In the model, r* is the equilibrium return on the only risk-free saving vehicle, traded among households. Argentina's negative ex-post peso rates mostly reflect surprise inflation, rate caps and financial repression, and households save largely in dollars and real estate rather than peso deposits. A target based on dollar returns would likely lie near 0–1%, and in that range Part I's conclusions apply. The thesis should state this choice explicitly and show both readings.

5. The informal premium is the piece of data that points to negative rates

Non-bank lenders in Argentina charge a real premium of 16–44 pp over banks, in line with the informal-credit literature. Holding ρ = 0.05, a premium of that size is what both models need to produce the 1993–2025 median, and what the 4-state model needs for the 1976–2025 median (Claim 12). It is not compatible with +2% or 0%. This argues for a negative target through a channel that does not rely on the ex-post peso rate. It does not by itself separate the models, and household debt points the other way: at the medians, the a_{min} route implies less debt than Argentina has ever had (Claim 11).

6. The policy-relevant difference survives every calibration

Each model calibrated to the same r*, the formalization counterfactual moves in the same direction in both, but the 4-state model predicts that financial constraints respond about one-third as much (Claim 6). That size disagreement, not a sign reversal, is the defensible contribution for policy evaluation.

What the thesis can and cannot claim after these tests

Claim	Status	Evidence
The 2-state model needs implausible parameters to match the observed real rate	Not supported at $r^* \geq 0$; supported only at the negative ex-post medians	Claims 3, 7, 9, 10
The 2-state model 'overestimates' r^*	Not supported: levels are equal by construction once both are calibrated	Claim 3
IH workers are what depresses r^*	Not supported: FH precautionary saving is the driver	Claim 4
Formalization reverses the sign of the borrowers-mass response	Not supported: it came from a non-equilibrium point	Claim 6
The two models imply different impatience at the same r^*	Supported (gap ≈ 0.024, robust)	Claims 3, 8
The 4-state model implies a much smaller response of credit constraints to formalization	Supported (about one-third)	Claim 6
Only the 4-state model can rationalise a negative real rate with $\rho > 0$	Supported, conditional on the ex-post peso rate being the right counterpart	Claims 9, 10
The 4-state model's credit requirements at +2% are realistic	Not supported: implied debt 17.7% vs 4.0%–12.3% observed	Claim 11
The thesis informal premium $\theta_I = 2$ pp is realistic	Not supported: 16–44 pp observed	Claim 12
Realistic informal premia are consistent with a negative r^*	Supported (both models at the 1993–2025 median, 4-state at the 1976–2025 median)	Claim 12

Table 19. Summary of claims.

Next steps: data that would settle the question

- **Calibrate with the observed premium.** Set θ_I to the data (about 35 pp) instead of 2 pp, and re-calibrate ρ to each target. Part III freed one parameter at a time; this joint calibration would show which ρ each model needs once the premium is realistic.
- **Direct data on informal borrowing rates.** Non-bank lenders are a proxy: some of their clients are formal, and moneylenders and store credit outside any registry are missing. A household survey with loan rates by employment status (ENGHo, the BCRA financial-inclusion survey) would measure θ_I directly.
- **Borrower shares**, formal and informal (Global Findex, ENGHo), to compare with the untargeted moments in Table 8 and Figure 9.
- **A dollar-based real return** for Argentine savers, to build a defensible positive r^* target.
- **Re-run the Case A/B/C reform comparison** (thesis Section 4.6) with each model at its own calibrated ρ: the welfare and revenue gaps in the thesis were computed at $\rho = 0.05$, where the 4-state economy at $\eta = 0.42$ did not clear.

6. Assumptions register

Every assumption the results depend on, where it enters, and how it was checked. 'Report' assumptions (R) were made in writing this document; all others come from the notebooks.

ID	Assumption	Where used	How checked / sensitivity
D1	Real rate is ex post: $r = (1 + i)/(1 + \pi) - 1$ with realised CPI inflation, not expected inflation.	Claim 1; Part II targets	Not checked against ex-ante measures; surprise inflation makes ex-post rates more negative.
D2	The 50-year series is the GMD short-term rate (CS2, Bordo, ECLAC, OECD, IMF IFS spliced), because Argentina has had no continuous policy rate.	Claim 1	BIS policy rate reported for 1993–2025 as a cross-check; also negative on average after 2002.
D3	Global Macro Database release 2026_09; forecast rows dropped.	Claim 1	–
D4	Data targets use medians, because the means are dominated by 1975–1991 high inflation.	Claims 9, 10	The 50-year mean (−11.1%) is far below the median. The 1993–2025 mean (−2.8%) is slightly above its median (−3.6%); it lies between the −2% and −3.56% cases, where the 4-state model already passes alone on ρ and a_{min} (Table 4, Figure 12).
D5	Household gross debt is BIS credit to households and NPISHs from all sectors (% of GDP, adjusted for breaks).	Claims 7, 11	An upper series scales it by 1 + 0.18 (BCRA: non-bank credit to individuals relative to bank credit, Feb 2026), in case BIS misses non-bank lenders.
D6	Model income (mean after-tax income) is measured by household consumption, % of GDP (World Bank). With bonds in zero net supply, model consumption equals income.	Claims 7, 11	Consumption was 62–70% of GDP over 1994–2025; using GDP instead would lower every ratio by about a third and strengthen the Claim 11 conclusion at +2%.
D7	The informal premium θ_l is measured by the personal-loan rate of non-bank credit providers over the bank rate.	Claim 12	Biased down: some non-bank clients are formal, and unregistered moneylenders are missing. Literature (Batini et al. 2010) reports informal rates often above 75% a year.
D8	Real premium = $(1 + i_{non\text{-}bank})/(1 + i_{bank}) - 1$, using nominal annual rates (TNA) from three BCRA reports.	Claim 12	TNA understate effective rates under monthly compounding, so the effective spread is larger. The spread includes default compensation (non-bank irregularity 34% vs 14% at banks, Feb 2026), which the model has no counterpart for.
C1	Exactly one parameter is freed per calibration; all others stay at thesis values (Section 2, $\eta = 0.42$).	All claims	Three alternative free parameters tested (ρ, a_{min}, θ_l).
C2	Thesis inputs used verbatim; 2-state $\gamma_l = 0.30$ (thesis text and Table 1) rather than 0.27 (notebook-4 code).	All claims	$\gamma_l = 0.27$ gives $\rho = 0.0498$ instead of 0.0542 (Table 6); the conclusions are unchanged.
C3	$r^* = +2\%$ is an external choice (advisor), not a data moment.	Part I	Targets 1–2.5% (Table 6) and 0% to −5.12% (Part II).
C4	r^* is the steady-state return on the only risk-free asset, in zero net supply among households, in a closed economy.	All claims	Not testable inside the model; it is the key caveat of Part IV, point 4.
C5	Thesis borrowing limit $a_{min} = -0.099$ (grid value, no stated source) and $\theta_l = 0.02$.	Claims 7, 10	Both freed in turn.

ID	Assumption	Where used	How checked / sensitivity
N1	Same solver as thesis notebooks 4 and 5 (implicit upwind finite differences, Δ = 1000, grid I = 800 on [−0.099, 5]).	All claims	Reproduces four thesis equilibria to < 5×10^{-7} (Table 3).
N2	For r ≤ 0: initial guess $V_0 = u(y + \rho a)/\rho$, warm starts, and a smaller step (Δ = 50) if needed.	Part II	Same fixed point: at r = 2% the value functions differ by 3×10^{-10}.
N3	A solve is invalid if the HJB does not converge or the density has negative mass; invalid points are never used.	All claims	Flagged in every scan.
N4	'Not attainable' means no sign change of S on the scan grid (a_{min} [−2, −0.01], θ_l [−0.05, 1]).	Claims 7, 10	Values outside these grids were not tested.
N5	Tight borrowing limits are resolved with few grid points below zero.	Claim 10	Re-calibrated at I = 2400: changes under 4% except one case at 8% (Table 12).
N6	A fixed-r root that does not clear when re-solved is treated as not attainable (2-state, θ_l, −5.12%).	Claim 10	S(r) is flat at −0.003 over r [−6.4%, −5.4%]; a knife-edge point.
N7	This report's figures come from a re-run of notebooks 8 and 8b on October 30, 2026, with MKL_THREADING_LAYER=SEQUENTIAL (the threaded MKL layer would not load in the execution environment).	All figures	Re-run CSVs compared with the previously committed CSVs; see the note below this table.
J1	ρ is plausible if it lies in [0.02, 0.12] per year (advanced-economy models use 0.02–0.06; emerging-market models use quarterly β ≈ 0.97–0.985).	Claims 3, 9, 10	The thresholds are editable (RHO_RANGE) in both notebooks.
J2	A negative informal premium $\theta_l < 0$ is not economically meaningful and fails.	Claims 7, 10	–
J3	A borrowing limit passes the feasibility rule whenever it exists and clears; its realism is judged separately.	Claims 7, 10	Judged against household-debt data in Part III (J4).
J4	A required a_{min} or θ_l is 'realistic' if the implied debt / income or the premium lies inside the observed range.	Claims 11, 12	The ranges are wide on purpose (lowest bank-only year to highest year with non-bank lenders; cheapest provider group to highest all-provider average). Two values sit near an edge: 2-state θ_l at −1% (15.2 pp vs 16 pp) and 4-state debt at −2% (5.0% vs 4.0%).
R1	Numbers in the text are read from the notebooks' result CSVs; a few printed-only values (Tables 3 and 12, r* at ρ = 0.05, the γ_l and η robustness) are transcribed.	Report	Transcribed values are marked in the captions.
R2	The negative ρ the 2-state model 'would need' at the medians extrapolates its constant impatience gap; it is not a solve.	Claims 8, 9	Figure 10 confirms S < 0 for every ρ ≥ 0.0025.
R3	Reading the FH exit rate as a frequent, large income drop interprets the estimated generator.	Claim 4	–
R4	That a dollar-based target would be near 0–1% is stated in notebook 8b but not computed from data in the repository.	Part IV, point 4	Unverified: flagged as a next step.
R5	Argentina's household credit is low.	Claim 7	Verified in Part III: 4.0%–10.4% of income since 1994 (BIS, D5–D6).

Reproducibility check (N7): after the first re-run (October 29, 2026), all 11 result tables that were already in the repository (data/interest_rates/*.csv) were compared with their committed versions. They are identical: same rows and labels, and the largest absolute difference in any number is 0. After adding the data benchmarks (October 30), the notebooks were re-run again and all 17 result tables matched the previous run exactly.

Files

Notebooks: *8. empirical_claims.ipynb* (Part I), *8b. empirical_claims_negative_r.ipynb* (Part II). Result tables: *data/interest_rates/*.csv*. Data benchmarks (raw files and sources): *data/interest_rates/benchmarks/*. Figures: *data/interest_rates/figures/*. This PDF is built by *9. empirical_claims_report.py*.

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